National Research University Higher School of Economics (HSE)

Curriculum

Field of study 01.03.02 Applied Mathematics and Information Science

Educational Programme "Data Science and Business Analytics"

Specializations: "Data Science in Applied Research", "Data Science in Business", "Data Science in Finance"

Implementing unit: Faculty of Computer Science, HSE - Moscow

4 th, 2026/2027 academic year

APPROVED

24.04.2026

Vice Rector

ROSHCHIN S.Y.

Signed with EDS

Length of Programme: 4 years

Years of Study: 2023/2024 - 2026/2027

Mode of Study: Full Time

Degree: Bachelor’s degree

Block Code	Course	Subject type	Department	Period of execution	Credits	Total Academic Hours	Contact Hours	Allocation of Contact Hours				Additional Information
								1	2	3	4	
	Degree Progamme				**60,00**	**2 280**	**460**	**128**	**126**	**200**	**6**	
	Final State Certification (FSC)				**3,00**	**114**	**2**				**2**	
	Thesis Work				**3,00**	**114**	**2**				**2**	
1	Presenting of Graduation Thesis	C			3,00	114	2				2A	Foreign language
	Internship				**15,00**	**570**	**6**	**2**			**4**	
	Research Internship				**15,00**	**570**	**4**				**4**	
1	Preparation of Graduation Thesis	C			9,00	342	2				2A	Foreign language
2	Graduation Internship	C		F	6,00	228	2				2A	Foreign language
	Professional Internship						**2**	**2**				
1	Work Experience Internship	C		F			2	2A				Foreign language
	Major				**42,00**	**1 596**	**452**	**126**	**126**	**200**		
	Elective Professional Block (Major)				**42,00**	**1 596**	**452**	**126**	**126**	**200**		
	Elective Courses				**14,00**	**532**	**152**	**56**	**56**	**40**		
	Elective Courses (Block 1)				**10,00**	**380**	**112**	**56**	**56**			
1	Introduction to the Study of Language for Computer Scientists	E	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Foreign language
2	Generative Models in Machine Learning	E	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Foreign language
3	Quantitative Finance	E	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Foreign language
4	Natural Language Processing	E	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Foreign language
5	Information Security Risk Management	E	Department of Big Data and Information Retrieval		5,00	190	56	28	28A			Foreign language
	Elective Courses (Block 2)				**4,00**	**152**	**40**			**40**		
1	Information Systems	E	Department of Big Data and Information Retrieval		4,00	152	60			60A		Online Course, Foreign language
2	Computer Vision	E	Department of Big Data and Information Retrieval		4,00	152	80			80A		Foreign language

3	Recommender Systems	E	базовая кафедра МТС		4,00	152	40			40A		Foreign language
	Compulsory Courses				**15,00**	**570**	**156**	**28**	**28**	**100**		
1	Statistical Methods for Market Research	C	Department of Big Data and Information Retrieval		6,00	228	60			60A		Foreign language
2	Strategy	C	Department of Big Data and Information Retrieval		9,00	342	96	28A	28	40A		Foreign language
	Specializations				**13,00**	**494**	**144**	**42**	**42**	**60**		
	Educational track "Data Science in Business"				**13,00**	**494**	**148**	**42**	**46**	**60**		
1	Core Management Concepts	C	Department of Big Data and Information Retrieval		9,00	342	96	28	28A	40A		Foreign language
2	Research Seminar "Data Analysis in Business 2“	C	Department of Big Data and Information Retrieval		4,00	152	52	14	18	20A		Foreign language
	Educational Track "Data Analysis in Applied Research"				**13,00**	**494**	**144**	**42**	**42**	**60**		
1	Classical and Neural Modeling	C	Department of Big Data and Information Retrieval		9,00	342	96	28	28A	40A		Foreign language
2	Research Seminar "Data Science in Applied Research 2"	C	Department of Data Analysis and Artificial Intelligence		4,00	152	48	14	14	20A		Foreign language
	Educational track "Data Science in Finance"				**13,00**	**494**	**148**	**42**	**42**	**64**		
1	Research Seminar "Data Science in Financial Markets"	C	Department of Big Data and Information Retrieval		4,00	152	52	14	14	24A		Foreign language
2	Asset Pricing and Financial Markets	C	Department of Big Data and Information Retrieval		9,00	342	96	28	28A	40A		Foreign language

Curriculum agreed:

Academic Supervisor	Voznesenskaia T.V.	19.03.2026
Dean	ARZHANTSEV I.V.	19.03.2026
Head of Centre for Educational Model Design	LEPESHKIN I.A.	23.04.2026

* Subject type:

Compulsory course	C
Elective course	E

Period of execution:

Fixed	F